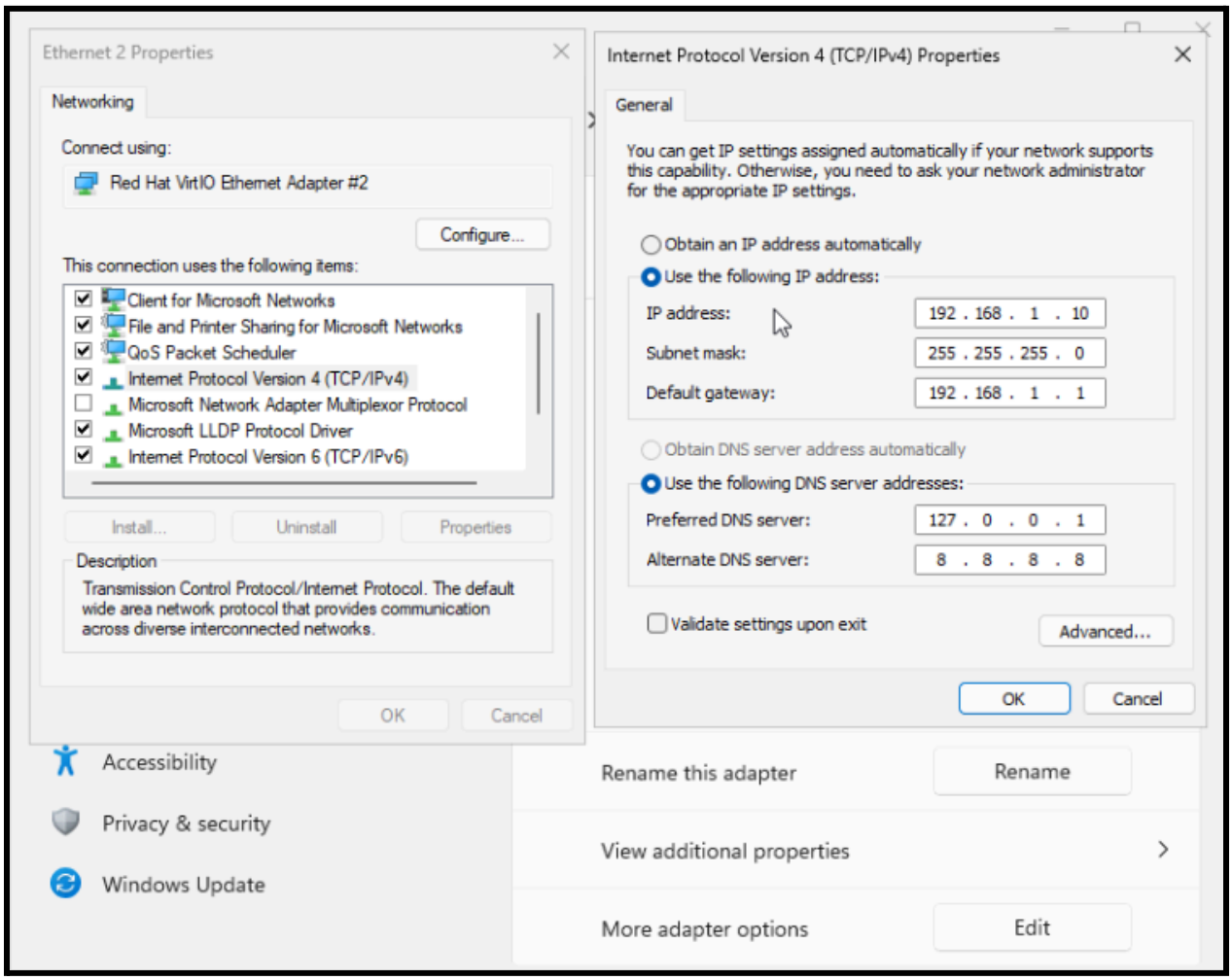
Network Configuration Summary

OS	IP	Name	Service
Vyos Router	WAN: 172.16.52.1 LAN: 192.168.1.1	N/A	Router
Windows Server 2025	192.168.1.12/24	DC1	Domain Controller 1
Windows Server 2016	192.168.1.19/24	WEB1	XAMPP Web Server
Windows Server 2012 R2	192.168.1.20/24	DC2	Domain Controller 2 IIS Web Server
Windows 10	192.168.1.16/24	WKS1	Client
Windows 7	192.168.1.14/24	SMB1	SMB
Ubuntu 24.04	192.168.1.17/24	SQL1	MySQL Database RocketChat Server
CentOS 10 (Coughlan)	192.168.1.13/24	WEB2	Apache Web
Debian 12	192.168.1.15/24	WEB3	NGINX Reverse Proxy
Ubuntu 25.04	192.168.1.18/24	FTP1	SFTP NextCloud

Setting up Windows Networks:

Windows 10 & 11:

- **Control Panel > View Network Status and Tasks > Change Adapter Settings > Ethernet > Properties > Internet Protocol Version 4**
- Set any IP address under your router's netmask, set the default gateway to your router's LAN IP.
- Preferably, set the DNS server to Google's public DNS server. Loopback could also work.



NOTE: Windows Server 2016 may have troubles connecting to the Internet. If that's the case, go back to Domain Controller 1 and run the commands:

- **Add-DnsServerForwarder 8.8.8.8**
- **Clear-DnsServerCache -Force**

This series of commands points DNS lookups to Google's public DNS server, and forces the cache's memory to be wiped to start clean instead of reusing answers from previous commands you inserted.

Setting up Linux Networks:

Ubuntu 25.04 (SFTP)

- **sudo chmod 600 /etc/netplan/50-cloud-init.yaml**
- **nano /etc/netplan/50-cloud-init.yaml**
- put the following information into the file:

```
network:
  version: 2
  ethernets:
    ens18:
      dhcp4: false
      addresses:
        - 192.168.1.18/24
      routes:
        - to: default
          via: 192.168.1.1
      nameservers:
         addresses: [1.1.1.1, 8.8.8.8]
```

Ubuntu 24.04 (MySQL, Roundcube) and Debian 12 (NGINX Reverse Proxy)

- **Settings > Wired Network Options (Cog Symbol) > IPv4**
- Input the following information:



CentOS 10 (Apache Webserver)

- **sudo chmod 600 /etc/NetworkManager/system-connections/ens18.nmconnection**
- Insert the following changes to the ens18.nmconnection file:

```
  GNU nano 8.1
[connection]
id=ens18
uuid=d51f7fb4-ed60-3380-93b6-79ad9e868970
type=ethernet
autoconnect-priority=-999
interface-name=ens18
timestamp=1759880710
autoconnect=true

[ethernet]

[ipv4]
method=manual
address1=192.168.1.13/24,192.168.1.1
dns=8.8.8.8;

[ipv6]
addr-gen-mode=eui64
method=auto

[proxy]
```

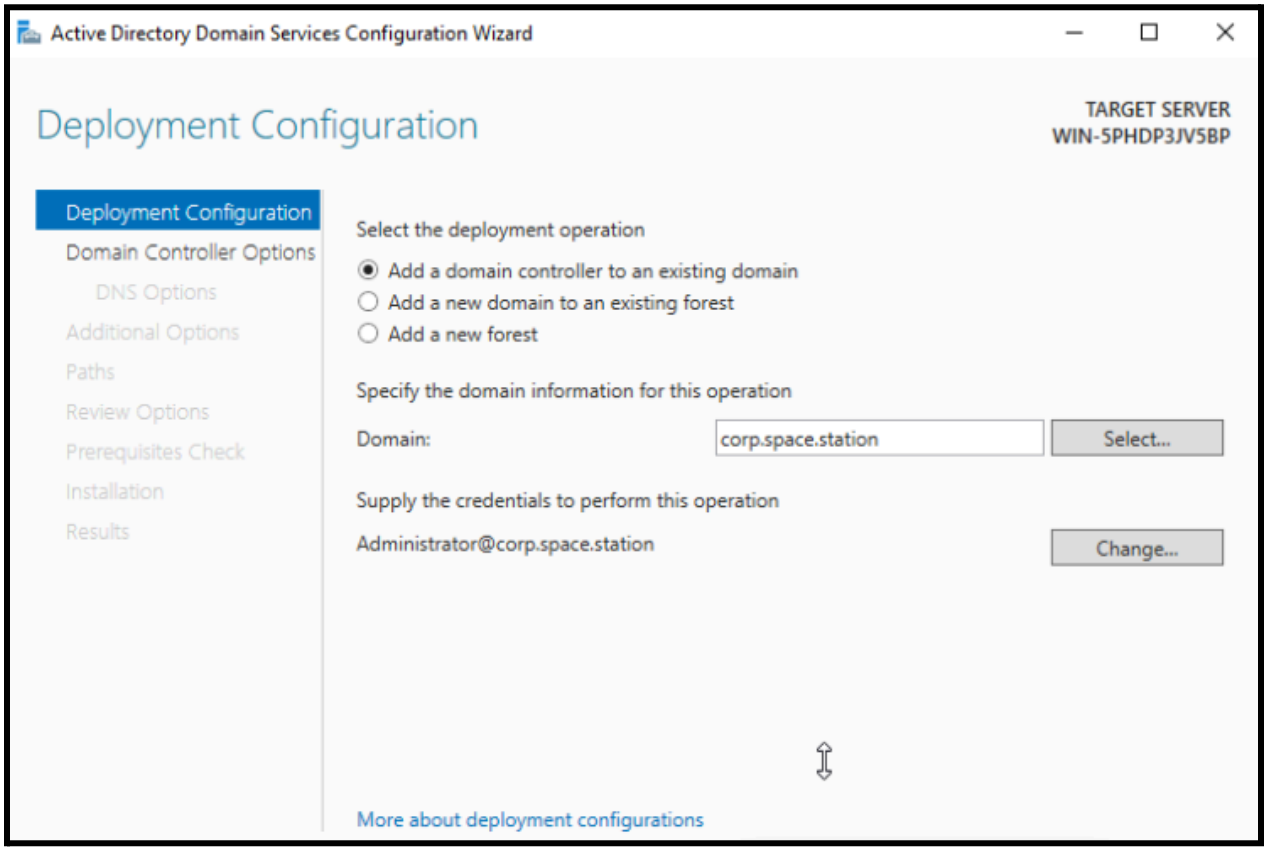
- Reload the connection using **sudo nmcli connection reload**
- Reboot the connection using **sudo nmcli connection up ens18**

```
PS C:\Users\Administrator> Get-Service adws, kdc, netlogon, dns | Select-Object Name, Status

Name      Status
----      ------
adws      Running
dns       Running
kdc       Running
Netlogon  Running


PS C:\Users\Administrator> Get-ADDomainController


ComputerObjectDN           : CN=WIN-KAR1BI0CIES,OU=Domain Controllers,DC=corp,DC=space,DC=statio
DefaultPartition           : DC=corp,DC=space,DC=station
Domain                     : corp.space.station
Enabled                    : True
Forest                     : corp.space.station
HostName                   : WIN-KAR1BI0CIES.corp.space.station
InvocationId               : fcb57ade-9efe-49a0-b839-7e31e987c45c
IPv4Address                : 192.168.1.10
IPv6Address                :
IsGlobalCatalog            : True
```



Domain Joining

Ubuntu 25.04:

- Go back to **sudo nano /etc/netplan/50-cloud-init.yaml** and change the nameservers from 8.8.8.8 to **BOTH** of your domain controller's IP addresses

```
      via: 192.168.1.1
  nameservers:
    addresses: [192.168.1.12, 192.168.1.20]
```

Ubuntu 24.04 & Debian 12:

- Go back to **Settings > Wired Network Options (Cog Symbol) > IPv4** and change the DNS IP addresses to the domain controller's IP addresses





CentOS 10:

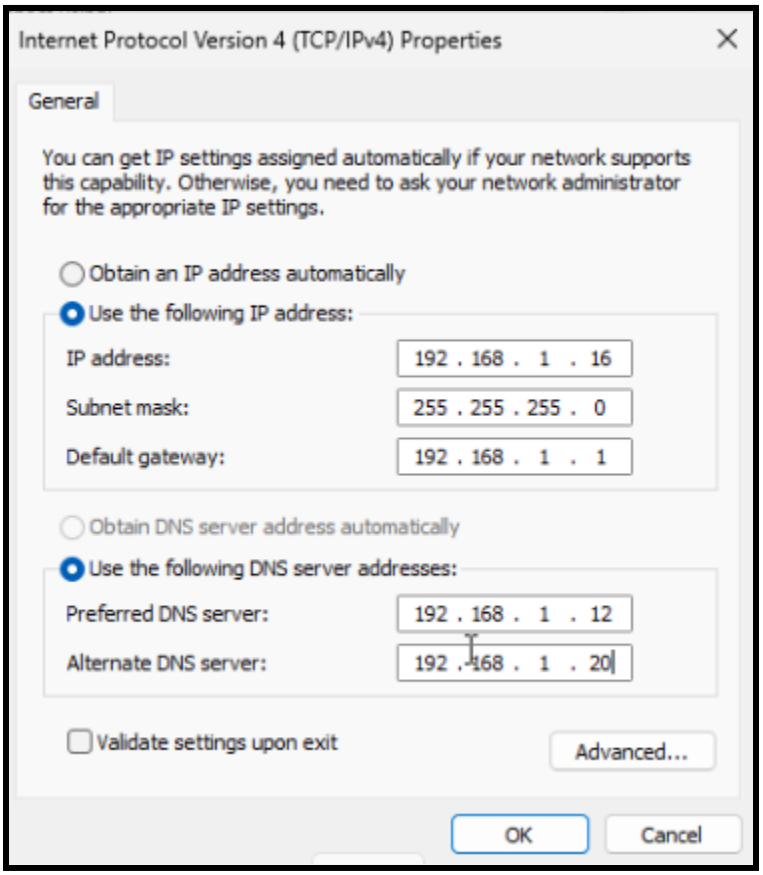
- Go back to **sudo nano /etc/NetworkManager/system-connections/ens18.nmconnections** and under **ipv4,** edit the **dns** variable to the IP addresses of the domain controllers.

```
[ipv4]
method=manual
address1=192.168.1.13/24,192.168.1.1
dns=192.168.1.12;192.168.1.20;
```

- Rerun **sudo nmcli connection reload**
- Rerun **sudo nmcli connection up ens18** (not sure if these are needed, but doing these just incase)

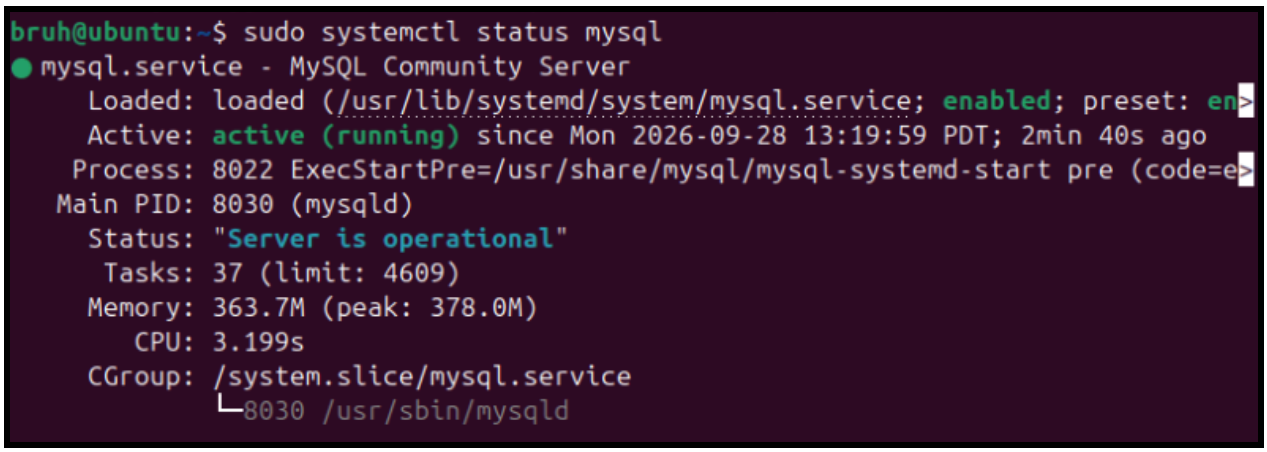
Windows Machines:

- **Control Panel > View Network Status and Tasks > Change Adapter Settings > Ethernet > Properties > Internet Protocol Version 4**
- Reconfigure the DNS server addresses to the two domain controller's IP addresses

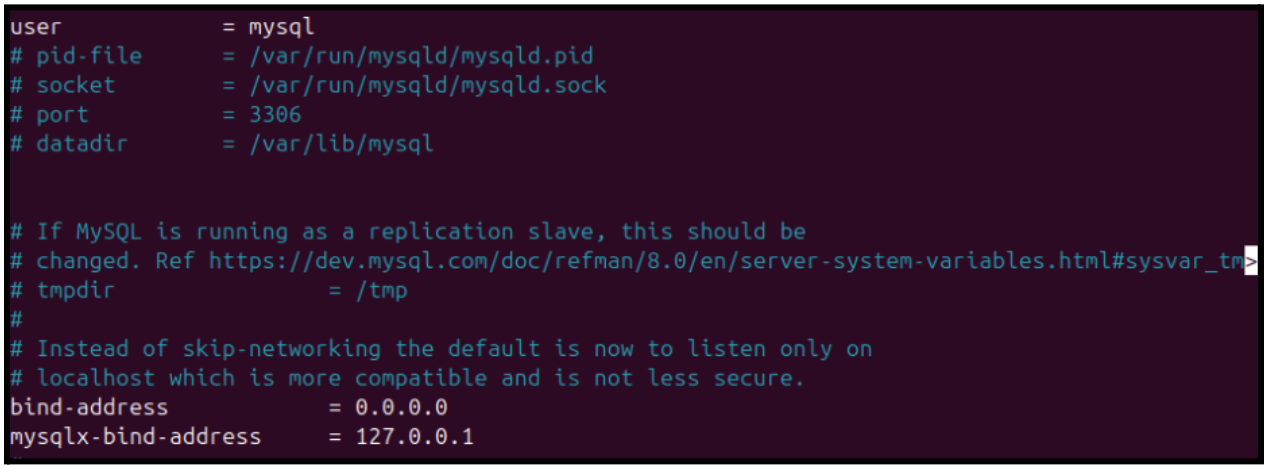


MySQL:

- Run **sudo apt-get update**
- Run **sudo apt install mysql-server -y** to install mySQL
- Run **sudo systemctl status mysql** to check if mySQL service is running, and **sudo systemctl start mysql** if not running yet.



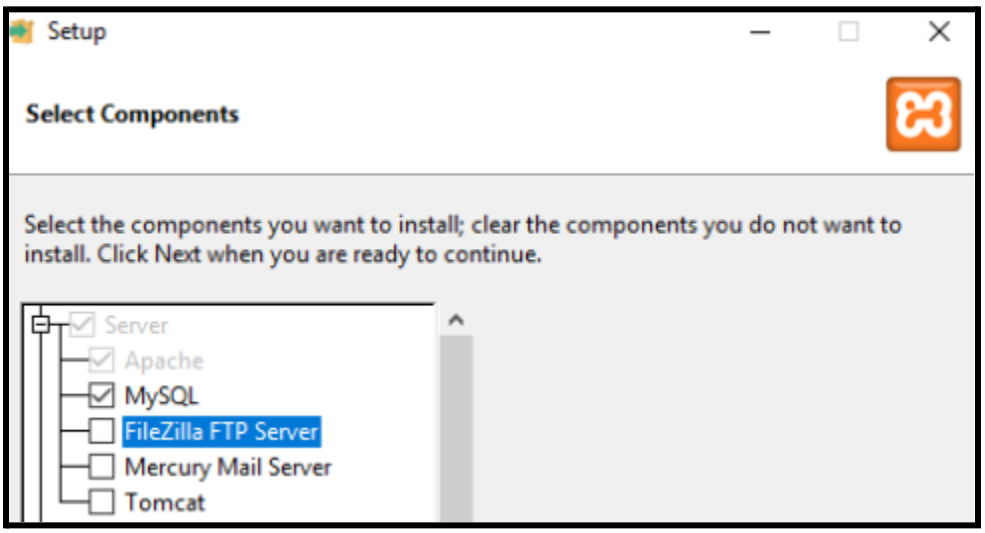
- Run **sudo sql_secure_installation**
  - During installation, enter **y** wherever prompted, and for password strength, select level 2.
- Run **sudo systemctl restart mysql** to ensure changes and installation are applied
- Run **sudo mysql -u root -p** and enter the password of your user account you're currently accessing the system with.
- To make sure other services can connect to the databases, edit the config file using **sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf** and change the bind-address to **0.0.0.0** to allow all machines to connect.



- We will need this SQL server for configuration in future parts of the setup, but leave it as it is for now.

XAMPP:

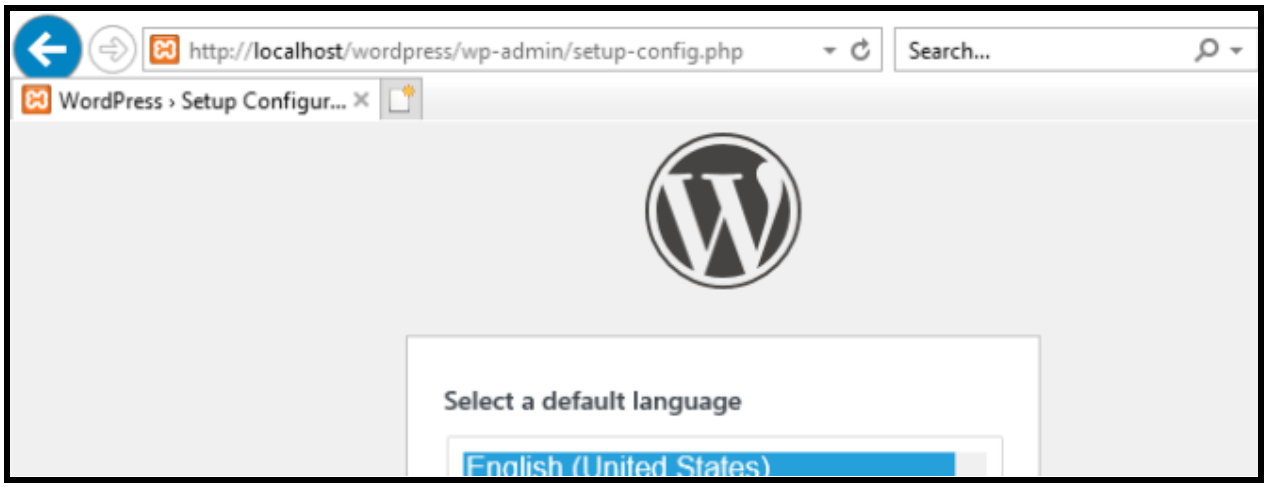
- **Server Manager > Local Servers > Internet Explorer Enhanced Security Configuration > Administrators: Off** for easier Internet access
- Go to https://www.apachefriends.org > **Downloads > Any Windows 64-bit Installation**
- In the installation wizard, uncheck **FireZilla FTP Server, Mercury Mail Server, and Tomcat**. Everything else can stay defaulted. Click **Next** till the installation proceeds.



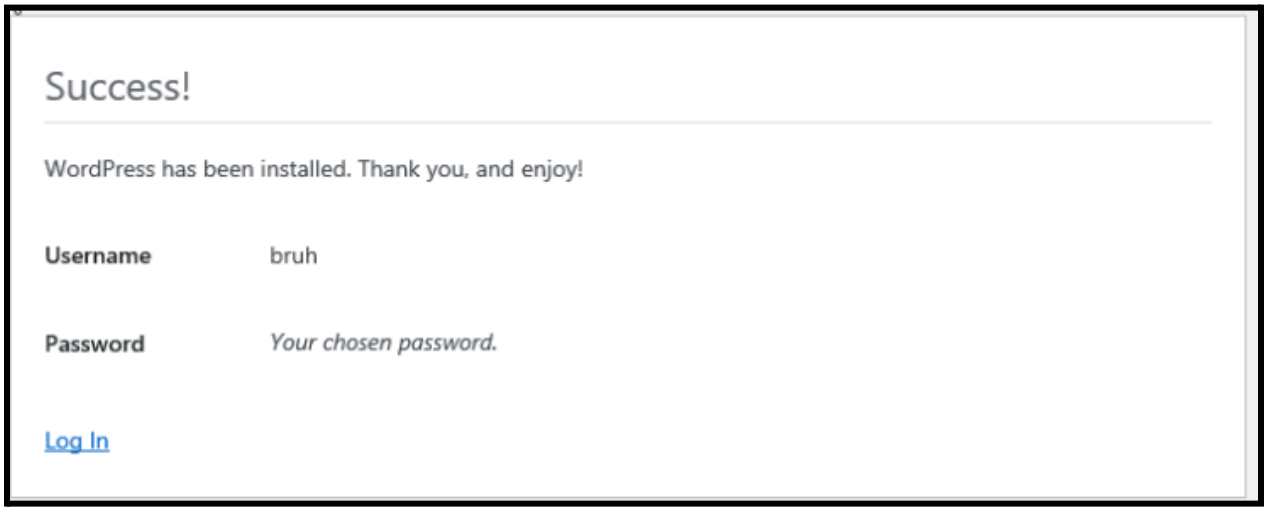
- Install WordPress at https://wordpress.org/download/ (latest version is fine)
- Extract the newly downloaded ZIP file to **C:\xampp\htdocs**
- In order to properly configure Wordpress, we'll need to create a database the website can access and a user associated with that database.
- To create a database, go back to the **Ubuntu 24.04 (MySQL)** machine > run **sudo mysql -u root -p** and enter your user accounts password > run **CREATE DATABASE {database-name};**
- To add a user, run **CREATE USER '{username}'@'%' IDENTIFIED BY '{password}';**

```
mysql> CREATE USER 'apache'@'%' IDENTIFIED BY 'ApacheCCDC2026!';
Query OK, 0 rows affected (0.03 sec)
```

- Grant privileges to the user with **GRANT ALL PRIVILEGES ON {database-name}.* TO '{username}'@'%';**
- Navigate to **http://localhost:80/wordpress/wp-admin** , or by clicking **Admin** on the XAMPP Control Panel on the Apache service.



- Connect to the database using the database you create on the MySQL machine, the username and password associated with that database, and the IP of the MySQL machine.
- Once prompted, click **Run the Installation**
- Enter the Site Title, Wordpress username, Wordpress password, and email that you plan on associating with your new Wordpress account.
- You will see the webpage below if everything works:



Apache:

- Updating everything by running **sudo yum update -y**
- Install Apache with **sudo yum install httpd -y**
- Start the Apache service using **sudo systemctl start httpd** and **sudo systemctl enable httpd**, and check the status of the service using **sudo systemctl status httpd**
- Create a new database for the Apache web server:
    - To create a database, go back to the **Ubuntu 24.04 (MySQL)** machine > run **sudo mysql -u root -p** and enter your user accounts password > run **CREATE DATABASE {database-name};**
    - To add a user, run **CREATE USER '{username}'@'%' IDENTIFIED BY '{password}';**

      ```
      mysql> CREATE USER 'httpd'@'%' IDENTIFIED BY 'httpdCCDC2026!';
      Query OK, 0 rows affected (0.04 sec)
      ```

    - Grant privileges to the user with **GRANT ALL PRIVILEGES ON {database-name}.* TO '{username}'@'%';**
- Next, install PHP
    - Run **sudo yum install php -y**
    - Run **sudo yum install php-mysqlnd -y**
    - Verify the installation with **php -v**
    - Restart Apache to load PHP with **sudo systemctl restart httpd**
- Next, Install WordPress
    - Use **wget https://wordpress.org/latest.tar.gz** to install wordpress
    - Use **tar -xzf latest.tar.gz** to extract the contents of the tar file you just downloaded
    - Move the wordpress files to **/var/www/html/** using **sudo mv ~/wordpress/ /var/www/html/**
    - Reconfigure file permissions so Wordpress has access
        - Apache user: **sudo chown -R apache:apache /var/www/html/**
        - Directories: **sudo find /var/www/html/ -type d -exec chmod 755 {} \;**
        - Files: **sudo find /var/www/html/ -type f -exec chmod 644 {} \;**
    - Make a directory for uploads using **mkdir /var/www/html/wp-content/uploads**
    - Copy and edit the config file:
        - **sudo cp /var/www/html/wordpress/wp-config-sample.php /var/www/html/wordpress/wp-config.php**
        - **sudo nano /var/www/html/wordpress/wp-config.php**

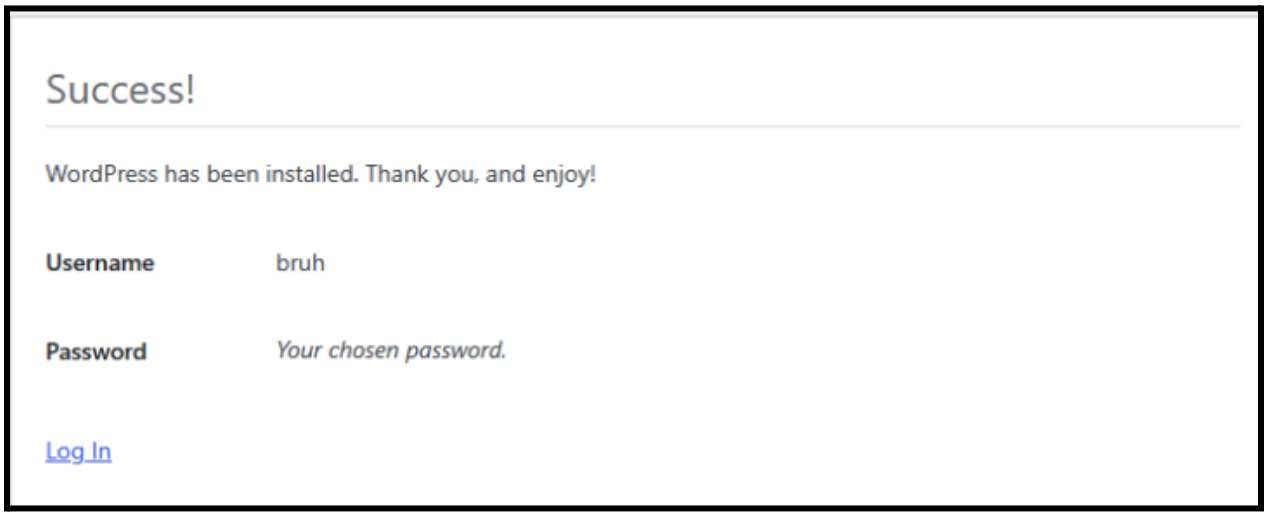
```
// ** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'httpd' );

/** Database username */
define( 'DB_USER', 'httpd' );

/** Database password */
define( 'DB_PASSWORD', 'httpdCCDC2026!' );

/** Database hostname */
```

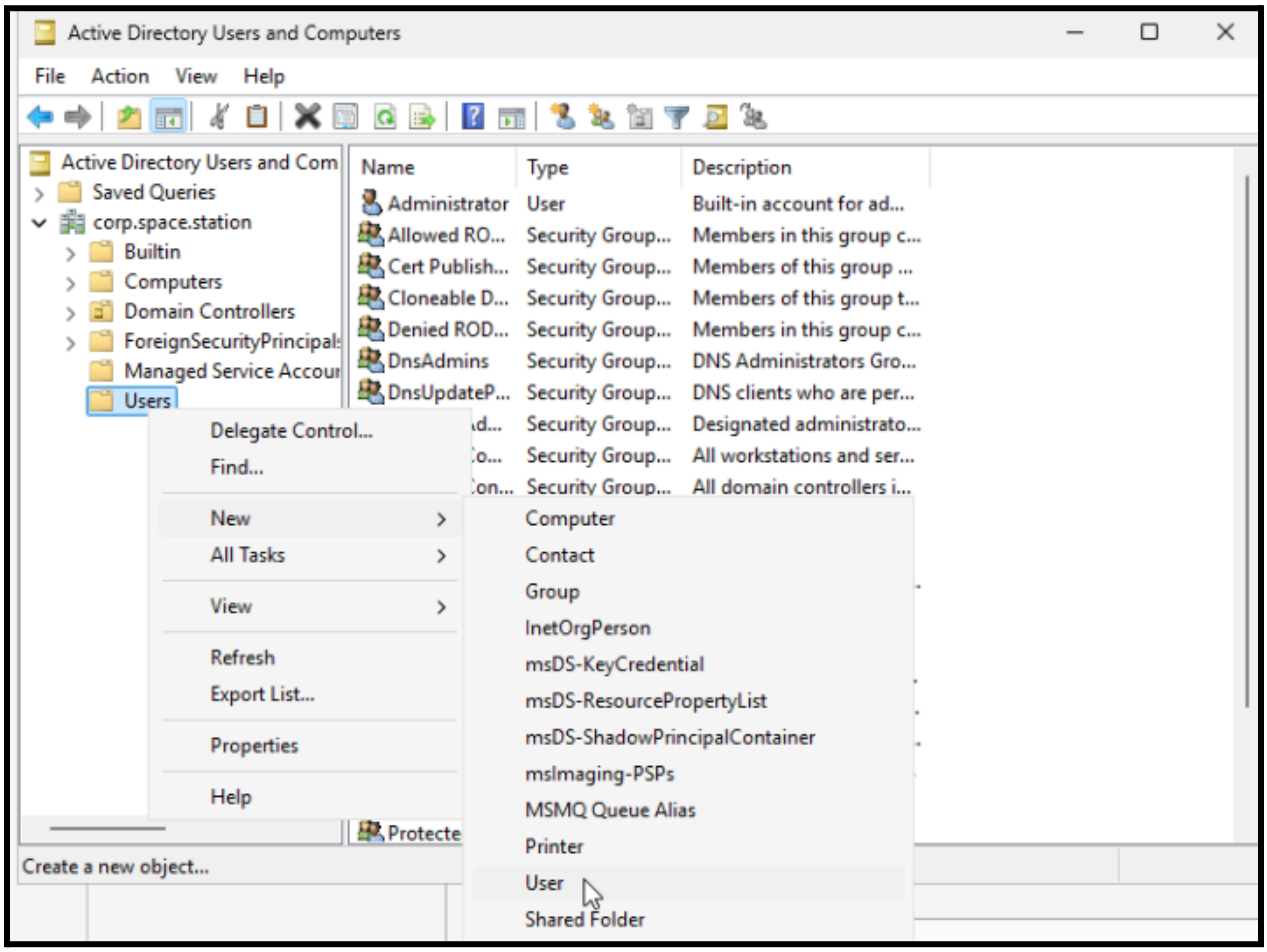
- Go to WordPress on a different machine via **http://{CentOS-External-IP}/wordpress** and fill out the Site title, username and password for the Wordpress site.



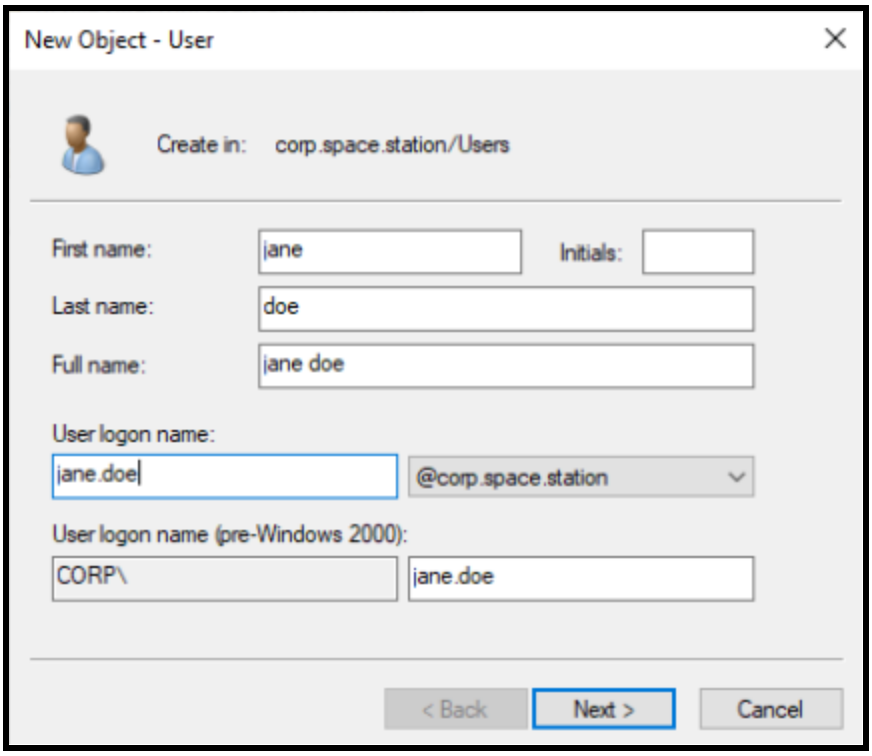
- Verify that the web server is ready by logging into the Wordpress site.

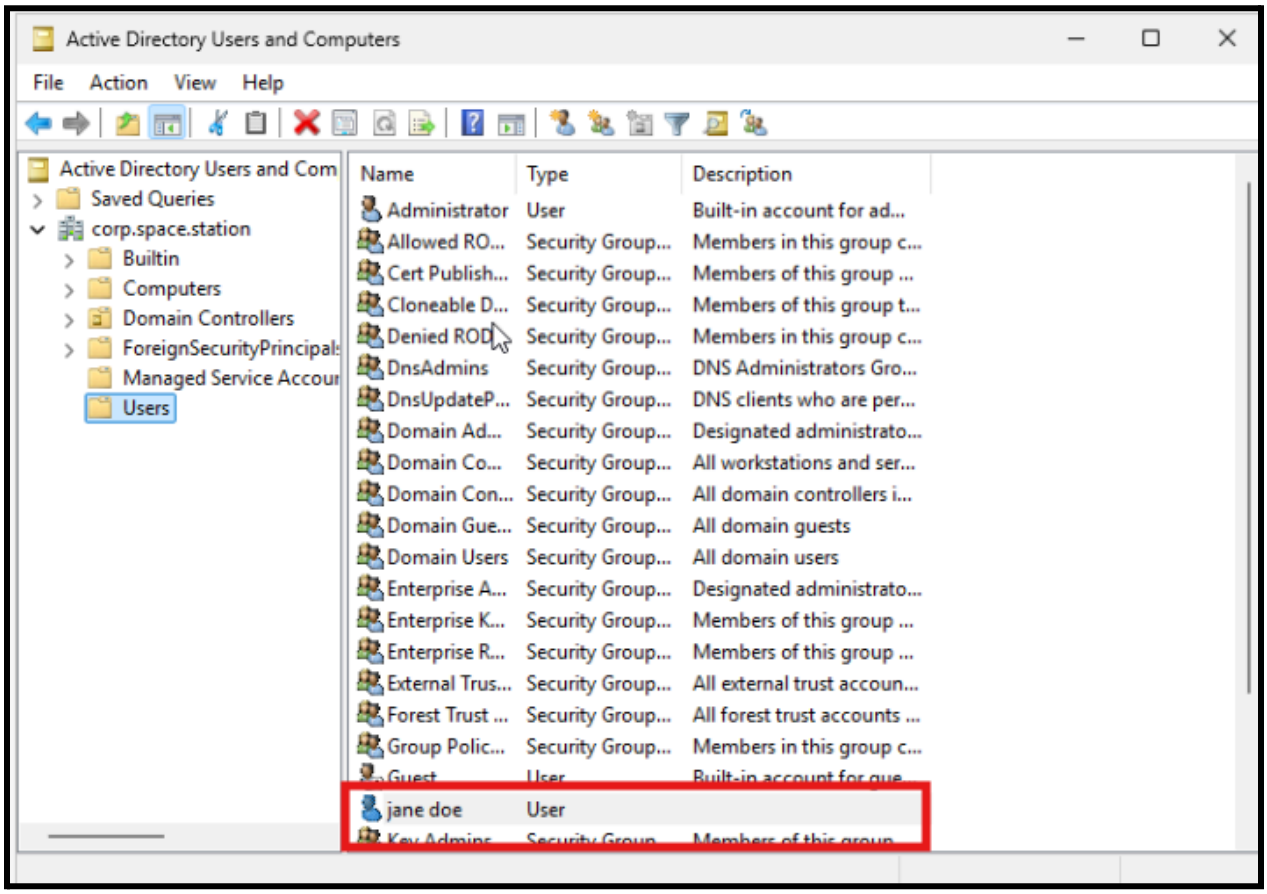
SMB:

- Go to **Server Manager > Tools > Active Directory User and Computers > [DOMAIN NAME] > Users > New > User**

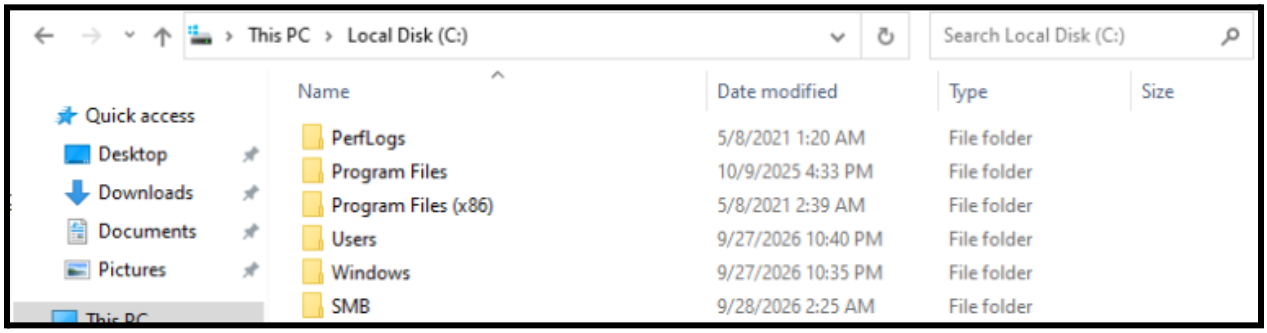


- Fill out all necessary information

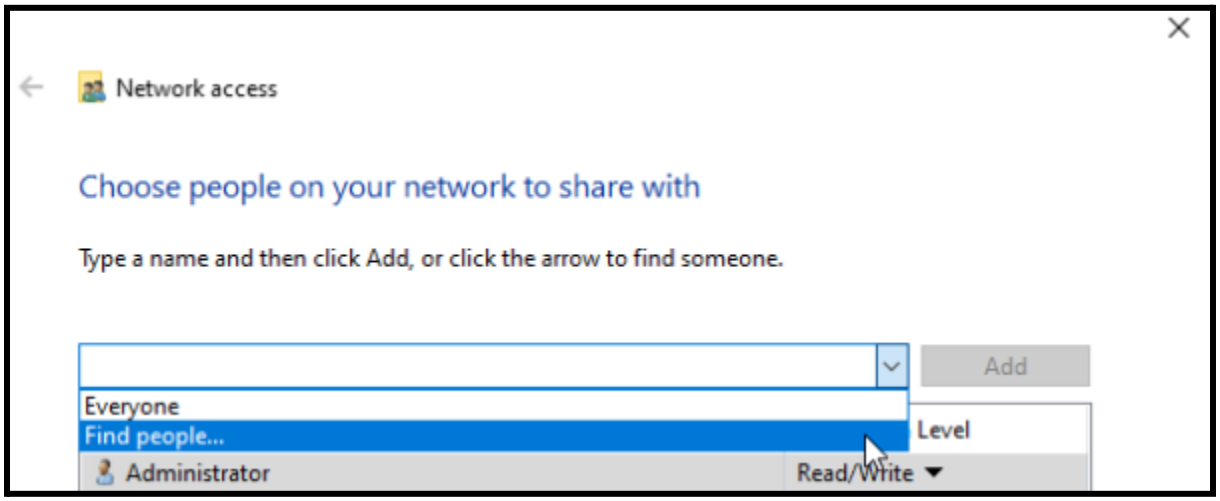

New Object - User
Create in: corp.space.station/Users
First name:
jane
Initials:
Last name:
doe
Full name:
jane doe
User logon name:
jane.doe
@corp.space.station
User logon name (pre-Windows 2000):
CORP\
jane.doe
< Back
Next >
Cancel


Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
corp.space.station
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accour
Users
Name Type Description
Administrator User Built-in account for ad...
Allowed RO... Security Group... Members in this group c...
Cert Publish... Security Group... Members of this group ...
Cloneable D... Security Group... Members of this group t...
Denied ROD... Security Group... Members in this group c...
DnsAdmins Security Group... DNS Administrators Gro...
DnsUpdateP... Security Group... DNS clients who are per...
Domain Ad... Security Group... Designated administrato...
Domain Co... Security Group... All workstations and ser...
Domain Con... Security Group... All domain controllers i...
Domain Gue... Security Group... All domain guests
Domain Users Security Group... All domain users
Enterprise A... Security Group... Designated administrato...
Enterprise K... Security Group... Members of this group ...
Enterprise R... Security Group... Members of this group ...
External Trus... Security Group... All external trust accoun...
Forest Trust ... Security Group... All forest trust accounts ...
Group Polic... Security Group... Members in this group c...
Guest User Built-in account for gue...
jane doe User
Key Admins Security Group... Members of this group

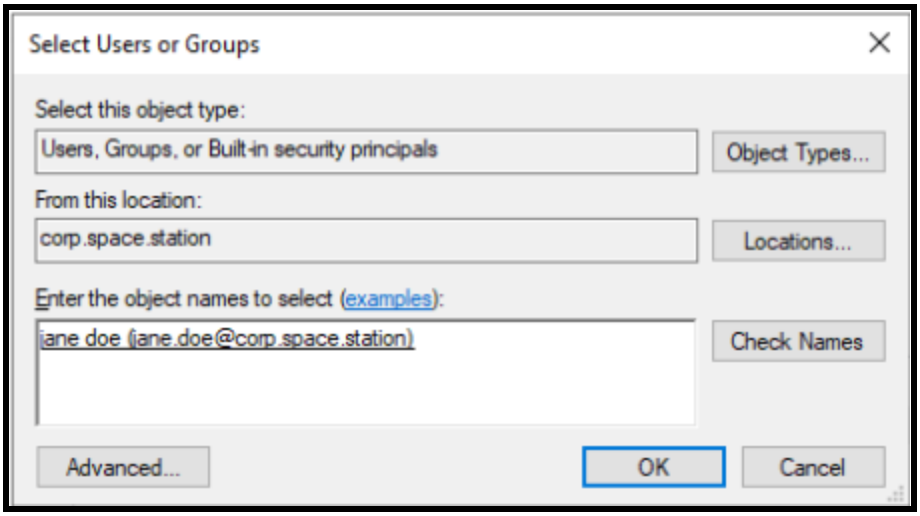
- In your file system, go to **This PC > Local Disk (C:)** and create a new file related to your SMB server

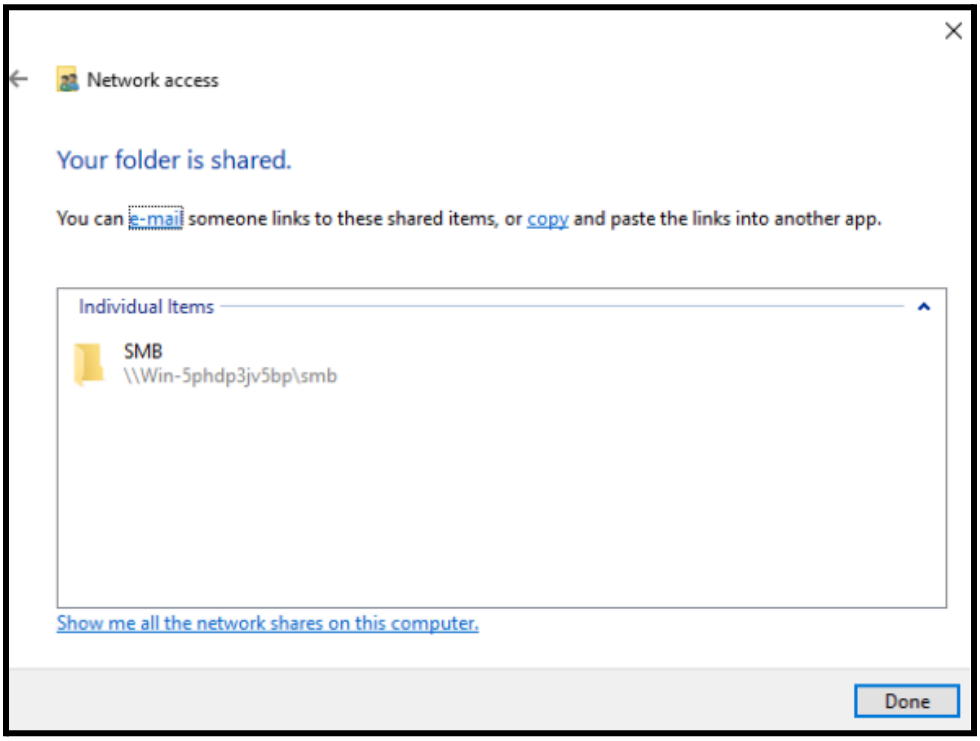


- Right-click on the newly created file and navigate to **Properties > Sharing > Share > Find People**

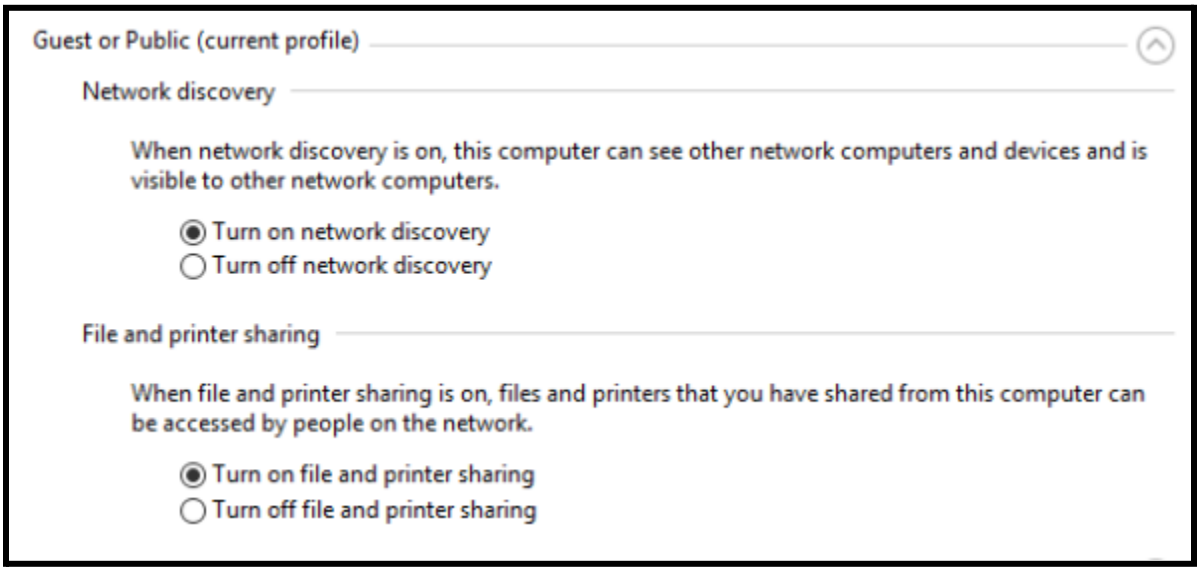


- In the text field, enter your newly created username and click **Check Name > OK**
- You will be redirected back to the dropdown menu, where you can press **Share**

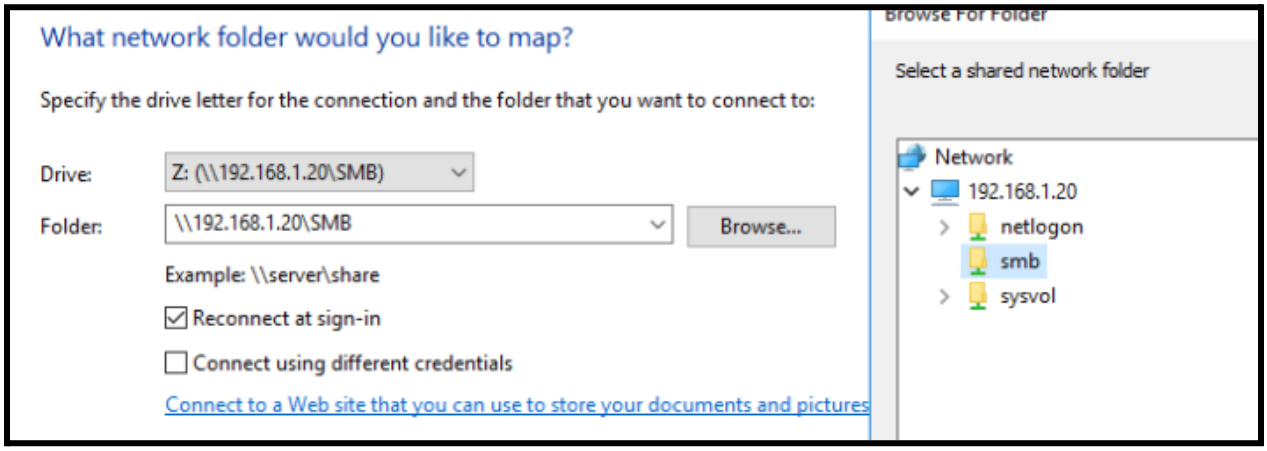




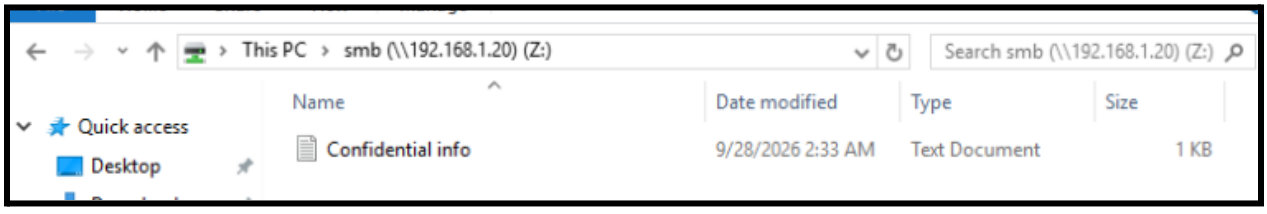
- On the Windows 10 machine, go to **Control Panel > Advanced Sharing Settings > Turn on network discovery + Turn on file and printer sharing**



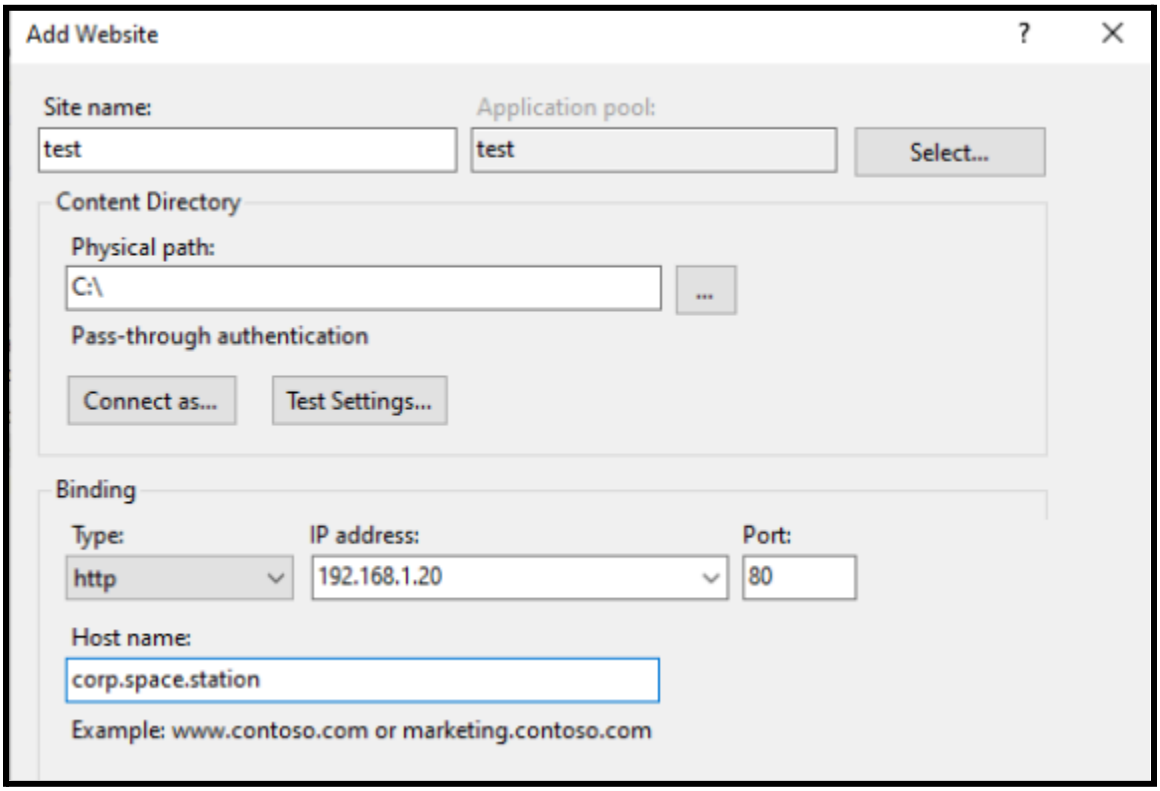
- Next, go to File explorer and right-click **This PC > Map network drive** and in the folder textfield, type in **\\{Domain Controller 2 IP}\{Shared File Name}**

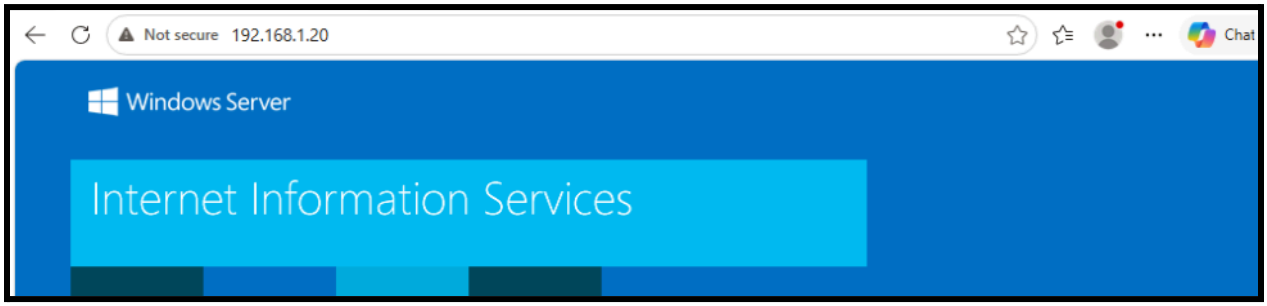


- Once you find the folder you created, click **OK,** and then enable **Connect using different credentials** and **Finish**. Enter the credentials for the created user account. You should be able to see the SMB share.



NGINX Web Proxy:

- Install the IIS Web Server
    - Go to **Server Manager > Add Roles and Features > Server Roles > IIS Web Server > Add Features**
        - Other settings can be left defaulted unless preferred otherwise
    - Click next until you reach **Confirmation**, then click **Install**
- 

- 

- Activate the NGINX Web Proxy
    - Go to the Debian 12 machine and use **su** to login to the root user
    - Run **sudo apt update && sudo apt upgrade** to ensure latest dependencies
    - Install NGINX with **sudo apt install nginx -y**
    - Start and verify the service is working with **sudo systemctl start nginx, sudo systemctl enable nginx, sudo systemctl status nginx**
    - Backup the default config file via **sudo cp /etc/nginx/sites-enabled/default /etc/nginx/sites-available/default-backup**

- Enter the following information into the **/etc/nginx/sites-enabled/default** file:

```
server {
        listen 80;
        listen [::]:80;
```

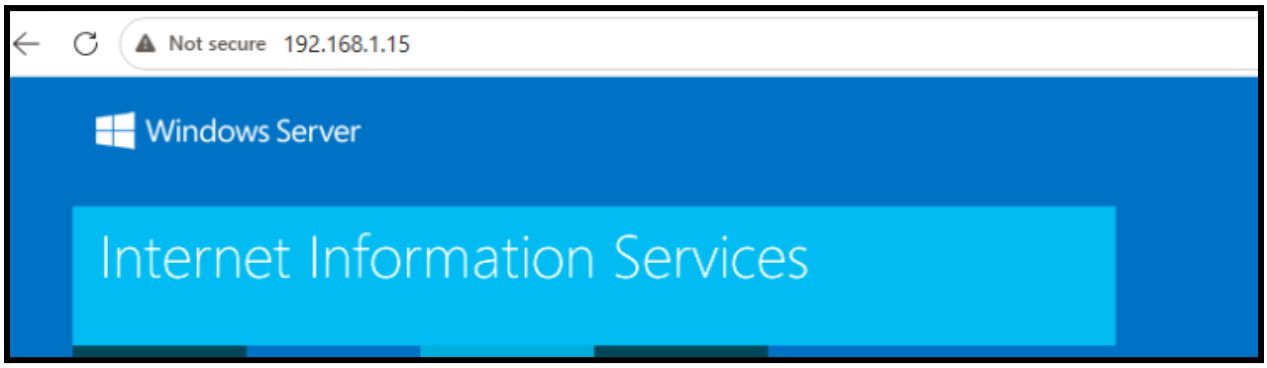
```
        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;

        server_name localhost;

        location / {
                # First attempt to serve request as file, then
                # as directory, then fall back to displaying a 404.
                proxy_pass http://192.168.1.20:80;
                proxy_set_header Host $host;
                proxy_set_header X-Real-IP $remote_addr;
                proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
                proxy_set_header X-Forwarded-Proto $scheme;
        }
```

- Run **sudo systemctl restart nginx** to ensure changes are applied
- Run **sudo nginx -t** to ensure syntax is valid
- Verify that the reverse proxy is working by accessing the web server via **http://{NGINX-IP}:80** on a different machine.



SFTP:

- Configure OpenSSH to support SFTP
    - Update the machine using **sudo apt update && sudo apt upgrade -y**
    - Install OpenSSH using **sudo apt install openssh-server -y**
    - Open the config file with **sudo nano /etc/ssh/sshd_config** and add/edit the file

```
# override default of no subsystems
Subsystem       sftp    internal-sftp

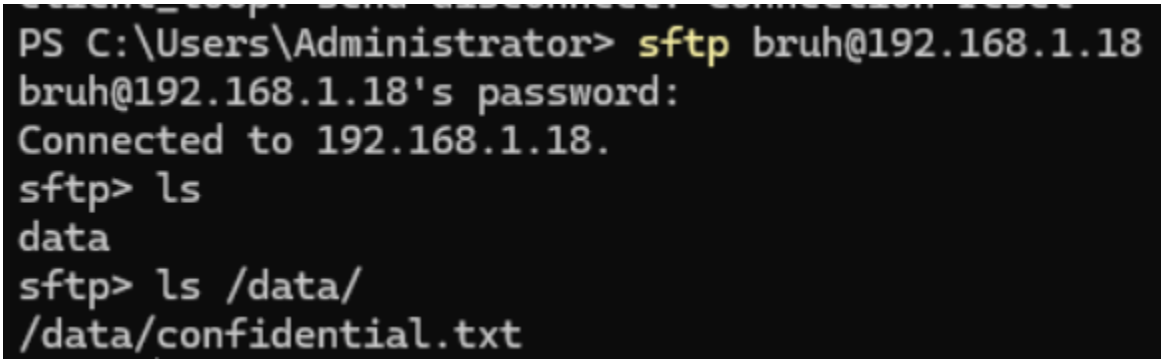
# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server

Match Group sftpusers
        X11Forwarding no
        AllowTcpForwarding no
        ForceCommand internal-sftp
        ChrootDirectory /srv/bruh
```

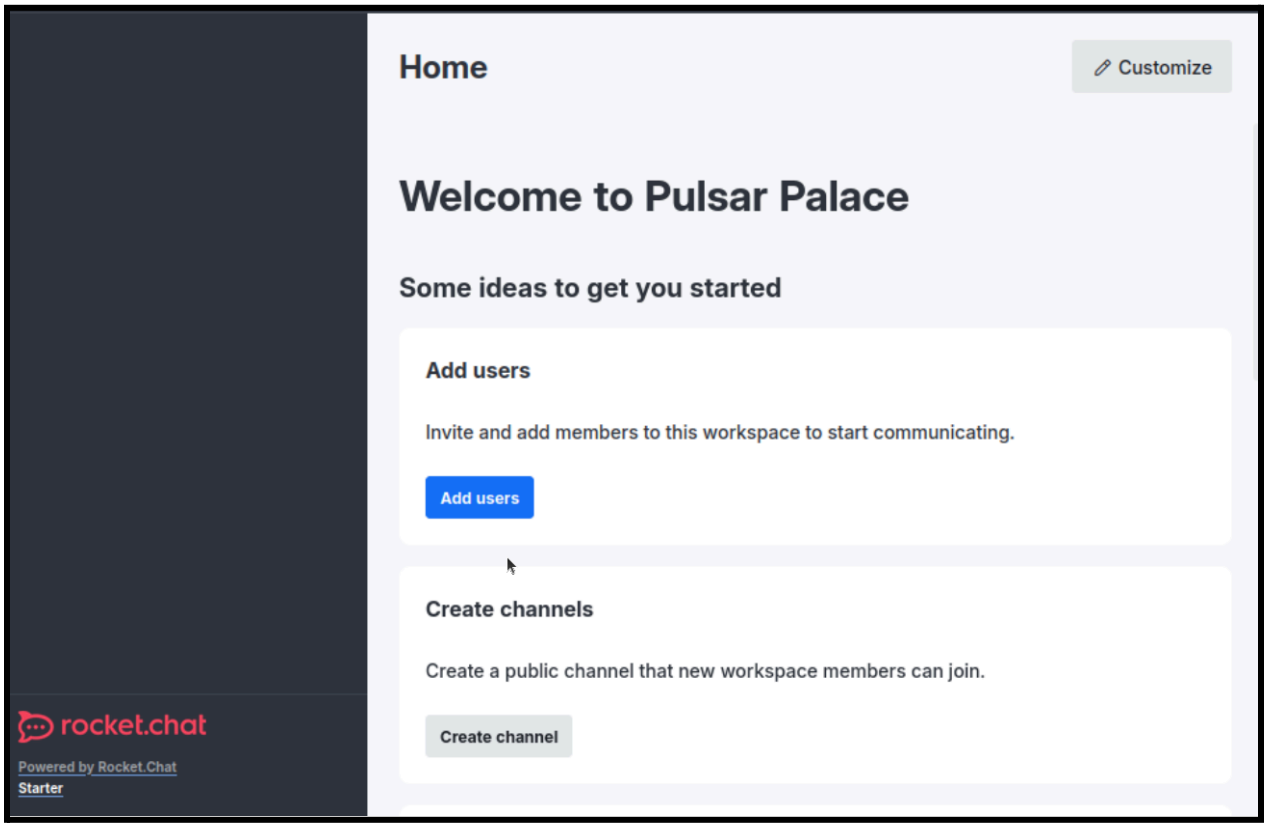
        - Comment out any tags that may counteract the settings above.

- Add SFTP users and group
    - To make a group, run **sudo groupadd {group name}**
    - To add a user to a group, run **sudo useradd -m -G {group name} -s /usr/sbin/nologin {username}**
    - Run **sudo passwd {username}** to create a new password for the SFTP user

- Create an SFTP directory to classified SWIFT info
    - Create a directory with **sudo mkdir -p /srv/{username}**
    - Edit perms with **sudo chown root /srv/{username}** and **sudo chmod g+rx /srv/{username}**
    - Create a directory in your ftp account with **sudo mkdir -p /srv/{username}/{dir name}**
    - Make your ftp user the owner of that directory with **chown {username}:{username} /srv/{username}/{dir name}**
- Verify that FTP service is running
    - Go to the windows 10 machine and run **sftp {SFTP username}@{SFTP IP}** and enter the password


PS C:\Users\Administrator> sftp bruh@192.168.1.18
bruh@192.168.1.18's password:
Connected to 192.168.1.18.
sftp> ls
data
sftp> ls /data/
/data/confidential.txt

Rocket.Chat (Ubuntu SQL machine):

- Run **snap install rocketchat.server**
- Go to localhost:3000
- Fill out all needed info, like username, email, etc
- Fill out organization info, also not really crucial
- Fill out admin email with something you'll remember, verify email
- 

- Check status with **systemctl status snap.rocketchat-server.rocketchat-server.service**

NextCloud (Ubuntu SFTP machine)

- Run **sudo snap install nextcloud**
- Go to SQL machine and add new user, database, and user privileges
  - **CREATE DATABASE 'nextcloud';**
  - **CREATE USER '[USERNAME]'@'%' IDENTIFIED BY '[PASSWORD]';**
  - **GRANT ALL PRIVILEGES ON nextcloud.* TO '[USERNAME]'@'%';**
- Download nextcloud on install page
  - **wget https://download.nextcloud.com/server/releases/latest.tar.bz2**
  - **wget** https://download.nextcloud.com/server/releases/latest.tar.bz2.sha256
- Unzip the nextcloud file
  - **Sudo apt install bzip2 -y**
  - **tar -xjvf [nextcloud tar file]**

    - **sudo cp nextcloud /var/www**
    - **sudo chown -R www-data:www-data /var/www/nextcloud**
- Configure the apache server for nextcloud
    - **sudo apt install apache2 libapache2-mod-php php-gd php-mysql php-xml**
    - **sudo nano /etc/apache2/sites-available/nextcloud.conf** and add the following configurations for the virtual host installation:

```
 GNU nano 8.3
<VirtualHost *:8080>
  DocumentRoot /var/www/nextcloud
  ServerName nextcloud.spacestation.com


  <Directory /var/www/nextcloud/>
    Require all granted
    AllowOverride All
    Options FollowSymLinks MultiViews

    <IfModule mod_dav.c>
      Dav off
    </IfModule>
  </Directory>
</VirtualHost>
```

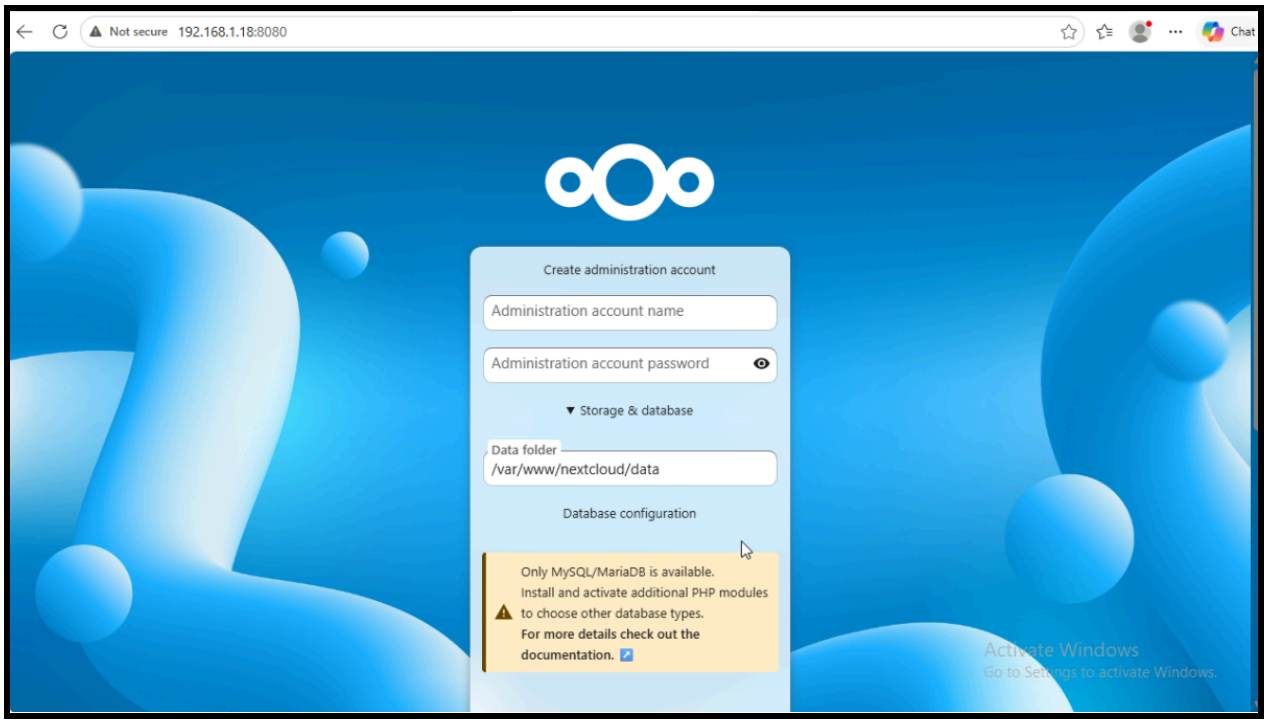
    - Modify **/etc/apache2/ports.conf** to listen to 8080:

```
Listen 8080

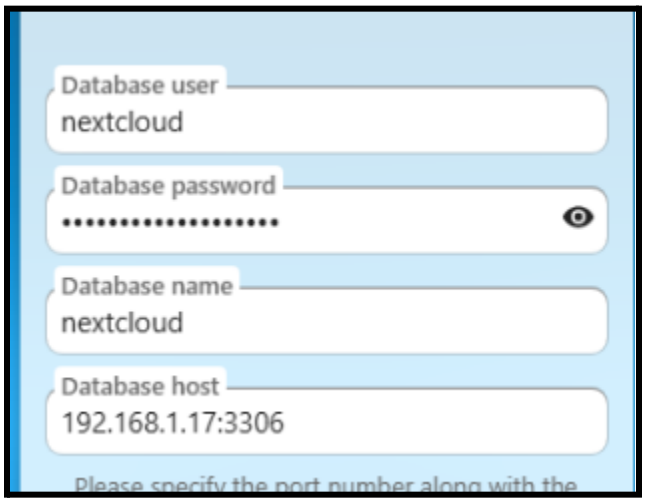
<IfModule ssl_module>
        Listen 443
</IfModule>

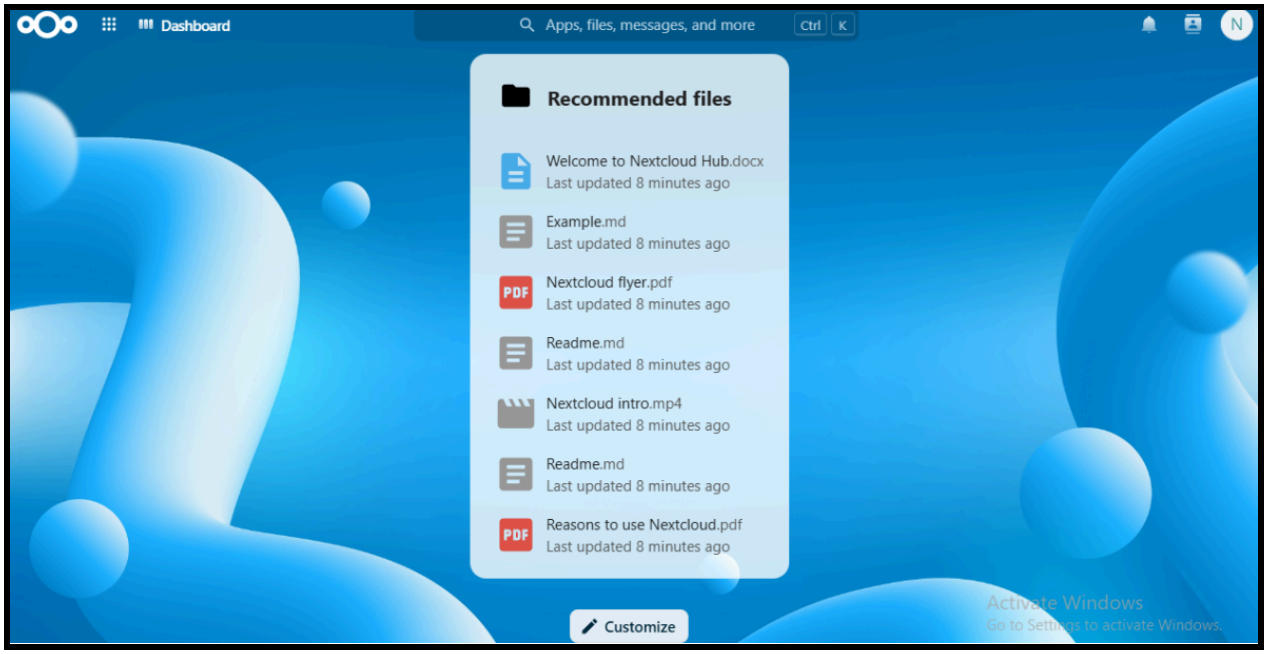
<IfModule mod_gnutls.c>
        Listen 443
</IfModule>
```

    - Run **sudo a2ensite nextcloud.conf**
    - Navigate to client machine and load up **192.168.1.[SFTP IP]:8080**



- Create nextcloud database
    - Fill out the information on your nextcloud page, including database name, user, password, and location that you configured on your mySQL machine.





- If you have issues with php, download the following addons:
  - **sudo apt install -y php-gd php-mysql php-curl php-mbstring php-intl php-xml php-zip php-bcmath php-gmp php-imagick php-apcu**

Security Considerations

1. **“SWIFT shouldn't be leaking out proprietary data to other clubs. That is how insurrections occur.”**
    a. Because the proprietary data is located on an SMB share only accessible by one user, the risk of the data getting leaked to other clubs is significantly reduced. To limit the access to only the Windows 10 client, run the following firewall commands on your SMB machine:
        - **New-NetfirewallRule -DisplayName “{Any Name}” -Protocol TCP -LocalPort 445 -RemoteAddress {Windows 10 IP} -Direction Inbound -Action Allow**
        - **Set-NetfirewallProfile -Profile Domain,Private,Public -Enabled True -DefaultInboundAction Block**
        - These firewalls allow inbound traffic from the Windows 10 machine and implicitly denies everything else.

2. **“SWIFT does not want violations of GDPR, as we operate worldwide. Make sure that HR data is locked down.”**
    a. The HR data can only be accessed by an SFTP account that is locked behind a group, username and password. Users are locked in a chroot directory so no other directories higher up can be accessed by the user. You can also establish firewalls to ensure that only the Windows 10 client is able to access SFTP.
        - **sudo iptables -A INPUT -i lo -j ACCEPT** to allow incoming requests from self
        - **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT** to allow existing traffic
        - **sudo iptables -A INPUT -p tcp –dport 22 -s {Windows 10 IP} -j ACCEPT**
        - **sudo iptables -P INPUT DROP** to deny anything else

3. **“The IIS web server shouldn't be reachable from any of its computer's interfaces. That's what the Web Proxy is for!”**
    a. Check **wf.msc** to verify that rules for all profiles implicitly allow outbound traffic and implicitly deny inbound traffic. Then go to **Inbound Rules** and sort by **LocalPort Ascending**. Look for any rules on **Port 80** and disabled if not needed. On Windows Powershell, run the command **New-NetFirewallRule -DisplayName “{Any Name}” -Profile Domain,Public,Private -Protocol TCP -LocalPort 80 -RemoteAddress 192.168.1.15 -Action Allow**

4. **“NSA released a different beast in 2017, however, you are running the same animal. Please name it.”**
    a. NSA was the creator of EternalBlue, a dangerous vulnerability that allowed for RCE on Windows systems via SMBv1 file sharing. SMBv1 is the animal that we are running on our Windows Machine. To disable it, run the command

Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol on any Windows Machines 2016 or older. If SMB is still needed, upgrade to SMBv2 or SMBv3

5. **WordPress creds** 🤔
    a. WordPress is notorious for having multiple vulns across different machines running it. Make sure that credentials for your Wordpress website follow a strong password hygiene to limit security risks of any brute force breaches occurring.

6. **We don't accept credit cards, but we still want to be 1/12th PCI-DSS compliant. Please follow PCI-DSS requirement 1 for all machines.**
    a. Requirement 1: **Install and maintain a firewall configuration to protect cardholder data**
    b. DC01: As the main Domain Controller, it doesn't need any additional firewall outside of the default Windows ones.
    c. DC02: As the secondary Domain Controller, default Windows firewalls work. Firewalls were created in relation to the IIS web server and NGINX reverse proxy as seen on **consideration 3**.
    d. Client: Implicitly deny all inbound requests, allow outbound requests to machines with **SFTP, SMB, NGINX Proxy, Apache server, and XAMPP server**.
        i. **Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True -DefaultInboundAction Block**
        ii. **New-NetFirewallRule -DisplayName "{Any Name}" -Protocol TCP -LocalPort {22/80/445} -RemoteAddress 192.168.1.{13/14/15/18/19} -DefaultInboundAction Allow**
    e. SMB: Machine should be receiving inbound requests from the client via port 445 and outbound requests to all ports on the 2 domain controllers.
        i. **New-NetfirewallRule -DisplayName "{Any Name}" -Protocol TCP -LocalPort 445 -RemoteAddress {Windows 10 IP} -Direction Inbound -Action Allow**
        ii. **Set-NetfirewallProfile -Profile Domain,Private,Public -Enabled True -DefaultInboundAction Block**
        iii. **New-NetfirewallRule -DisplayName "{Any Name}" -RemoteAddress {DC 1 & 2 IP} -Direction Outbound -Action Allow**
        iv. **Set-NetfirewallProfile -Profile Domain,Private,Public -Enabled True -DefaultOutboundAction Block**
    f. Apache: Allow inbound requests on port 80 from the client machine, and outbound requests to port 3306 to the SQL database
        i. **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
        ii. **sudo iptables -A OUTPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
        iii. **sudo iptables -A INPUT -p tcp –dport 80 -s {Client IP} -j ACCEPT**
        iv. **sudo iptables -A OUTPUT -p tcp –dport 3306 -s {SQL IP} -j ACCEPT**

v. **sudo iptables -P OUTPUT DROP**
vi. **sudo iptables -P INPUT DROP**

g. XAMPP: Allow inbound requests on port 80 from the client machine, and outbound requests to port 3306 to the SQL database
   i. **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   ii. **sudo iptables -A OUTPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   iii. **sudo iptables -A INPUT -p tcp –dport 80 -s {Client IP} -j ACCEPT**
   iv. **sudo iptables -A OUTPUT -p tcp –dport 3306 -s {SQL IP} -j ACCEPT**
   v. **sudo iptables -P OUTPUT DROP**
   vi. **sudo iptables -P INPUT DROP**

h. NGINX Proxy: Allow inbound requests on port 80 from the client machine, and outbound requests to port 80 to the secondary Domain Controller
   i. **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   ii. **sudo iptables -A OUTPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   iii. **sudo iptables -A INPUT -p tcp –dport 80 -s {Client IP} -j ACCEPT**
   iv. **sudo iptables -A OUTPUT -p tcp –dport 80 -s {DC 2 IP} -j ACCEPT**
   v. **sudo iptables -P OUTPUT DROP**
   vi. **sudo iptables -P INPUT DROP**

i. MySQL: Allow inbound requests on port 3306 from the XAMPP and Apache web servers, don’t allow any outbound requests.
   i. **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   ii. **sudo iptables -A OUTPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   iii. **sudo iptables -A INPUT -p tcp –dport 3306 -s {XAMPP IP} -j ACCEPT**
   iv. **sudo iptables -A INPUT -p tcp –dport 3306 -s {Apache IP} -j ACCEPT**
   v. **sudo iptables -P OUTPUT DROP**
   vi. **sudo iptables -P INPUT DROP**

j. Rocket.Chat: Allow inbound requests from the XAMPP and Apache web servers. Output is needed on the local machine to access the website locally.
   i. **sudo iptables -A INPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   ii. **sudo iptables -A OUTPUT -m conntrack –ctstate ESTABLISHED,RELATED -j ACCEPT**
   iii. **sudo iptables -A INPUT -p tcp –dport 3300 -s {XAMPP IP} -j ACCEPT**
   iv. **sudo iptables -A INPUT -p tcp –dport 3000 -s {Apache IP} -j ACCEPT**
   v. **sudo iptables -P OUTPUT DROP**
   vi. **sudo iptables -P INPUT DROP**
   vii. **sudo iptables -A OUTPUT -o lo -j ACCEPT**

Sources Used:

https://linuxvox.com/blog/how-to-install-wordpress-with-apache-on-centos-7/
- Installing and Configuring Wordpress & Apache on CentOS

https://linuxize.com/post/how-to-install-mysql-on-ubuntu-24-04/
- Installing and Configuring MySQL on Ubuntu

https://learn.microsoft.com/en-us/iis/install/installing-iis-85/installing-iis-85-on-windows-server-2012-r2
- Installing and Configuring IIS server on Windows 2012 R2

https://linuxconfig.org/setting-up-nginx-reverse-proxy-server-on-debian-linux
- Establishing NGINX Reverse Proxy to pair with IIS server

https://sftpcloud.io/learn/sftp/ubuntu-sftp-server-how-to-install-and-configure
- Installing and Configuring SFTP server on Ubuntu

https://learn.microsoft.com/en-us/windows-server/storage/file-server/troubleshoot/detect-enable-and-disable-smbv1-v2-v3?tabs=server
- Fighting back against EternalBlue

https://www.isms.online/pci-dss/requirement-1/
- PCI-DSS requirement 1

The big clawde
- Weird bugs/errors that I had to troubleshoot

https://ubuntu.com/tutorials/setup-rocketchat-server-on-ubuntu#5-changing-port-and-enabling-ssl
- Rocket.chat

https://docs.nextcloud.com/server/stable/admin_manual/installation/source_installation.html#apache-configuration-label

- Nextcloud